

What motivates academics in Egypt toward generative AI tools? An integrated model of TAM, SCT, UTAUT2, perceived ethics, and academic integrity

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Abstract

In recent years, the adoption of AI technologies in academia has increased, prompting a need to explain factors driving scholars to adopt or plan to adopt AI technologies in research routines. This study integrates three models into one integrated model: TAM, UTAUT, and SCT. These three models are combined to understand how GenAI self-efficacy, perceived ethics, academic integrity, social influence, facilitating conditions, perceived risks, ease of use, and perceived usefulness influenced the participants' intention to adopt GenAI in research. Following this, data were collected from Egyptian academics linked to Egyptian universities. There were 742 responses to this question. Data were analyzed using Partial Least Squares Structural Equation Modelling (PLS-SEM). The paper's results showed that GenAI self-efficacy, social influence, and perceived ethics are significantly related to perceptions of ease of use, usefulness, and intention to use GenAI. Facilitating conditions have a negative effect on perceived ease of use, and perceived risk does not affect perceived usefulness or intention to use significantly. Notably, the result found that ethics and academic integrity affect perceptions of GenAI's usage and utility. This guide illustrates how universities must take proactive steps to influence how AI will be used and reinforces the importance of using these tools within an ethical lens. This paper emphasizes the need to balance the use of generative AI and ethical practices. It examines the role of ethics and academic integrity in perceptions that influence attitudes toward the use of generative AI. They represent a step forward in our understanding of how to induce AI adoption—in this case, in the academic context, and specifically in Egypt. Additionally, it places sound emphasis on the role of AI technology and how it can be beneficial whilst advocating for a sensible approach in that application, which includes ethical principles.

Keywords

generative AI, Egyptian academics, TAM, UTAUT2, SCT, academic integrity, perceived ethics, perceived risks, ChatGPT, technology adoption

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Introduction

Today's AI tools, especially natural language processing (NLP) tools like ChatGPT and Gemini, are now a part of how academic research and education get done.

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They provide more efficiency, innovation, and accessibility (Bin-Nashwan et al., 2023). Generative AI tools can help with idea generation, writer's block, task automation, and content generation (Eke, 2023; Strzelecki and ElArabawy, 2024). Nonetheless, they also pose an ethical challenge, from academic fraud and plagiarism to diminished originality and producing deceptive, misleading content that threatens academic integrity (Acosta-Enriquez et al., 2024). Similarly, ownership of AI-generated content poses intellectual property concerns. (Bouteraa et al., 2024).

With the increasing concerns about using AI in academia, some universities have banned tools like ChatGPT to avoid unethical use of AI among students and researchers. Still, that sort of measure does not address the underlying ethical issues of AI. Experts recommend bringing training on the ethics of AI technologies into curricula at schools and universities, focusing on transparency, fairness, and accountability of such systems (Eke, 2023). Responsible AI adoption is mediated by a framework like the Multiscale Ethics Framework, which can help ensure the AI operates within the constraints of privacy and integrity. Striking a balance would help realize the full potential of GenAI while minimizing risks and promoting responsible innovation in academia (Acosta-Enriquez et al., 2024).

Many recent studies have attempted to study the application and influence of GenAI in academia, as well as other problems, such as the challenge and perceptions of the academic community. (Cotton et al., 2024; Ivanov et al., 2024). However, there is a limited understanding of perceived ethics, academic integrity, and psychological and technological factors and how they influence user behavior. Moreover, the literature reveals a lack of studies that focus on the motivations of Egyptian researchers to use GenAI tools. Hence, our study aims to fill this gap by examining the factors that shape the use of GenAI tools among Egyptian researchers.

Combining the Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT2), and Social Cognitive Theory (SCT) provides a valuable perspective and integrated model for understanding user behaviour towards adopting new technology. We have extended these models by integrating measures of academic integrity, perceived ethics, and their influence on academic and higher education researchers' intentions in utilizing GenAI tools. In the area of employing GenAI for research, this study: (1) Examines the impact of GenAI self-efficacy, perceived ethics, academic

integrity, social influence, facilitating conditions, and perceived risks on the ease of use and perceived usefulness of GenAI in scientific research; (2) Investigates how GenAI self-efficacy, perceived ethics, academic integrity, social influence, facilitating conditions, perceived risks, ease of use, and perceived usefulness affect the intentions to utilize GenAI tools in scientific research. Our findings should help academia better understand the use of generative AI tools, alongside the academic integrity challenges they pose. The findings support higher education institutions in developing stronger frameworks and policies that leverage the advantages of GenAI tools while minimizing the risks surrounding these technologies in academic research and teaching.

Literature review and hypotheses development

Generative AI in academic contexts

The literature reveals an increasing interest in adopting GenAI in academia, research, and knowledge generation (Kshetri et al., 2023). One of these studies suggested that GenAI tools can change the educational and research landscape by serving as educational and research tools that work alongside humans in providing knowledge (Ivanov et al., 2024). This changed the current practices in educational institutions from a "teacher-student" to a "teacher-AI-student" model, allowing innovative learning and research experiences (Foroughi et al., 2024).

GenAI holds transformative promise in education and research, with the need to discuss how these tools should be used rather than fear the potential for disruption (Susarla et al., 2023). GenAI models stand out in other work, such as statistical process control, education, and research (Megahed et al., 2024). Yet, as these technologies are still developing, they are susceptible to misuse and, hence, require extensive empirical studies to understand their meanings in educational and research environments.

The use of GenAI in academic writing was explored by many scholars. Based on the literature, it improves efficiency and quality with concept clarification, language translation, personalized feedback, and stylistic changes (Lingard, 2023). It further facilitates structural, dialogical, and linguistic enhancement, which is beneficial for completing tasks such as topic generation, outlining, content revision, and reflective writing (Su et al., 2023). ChatGPT helps solve writer's block and

automates repetitive writing tasks (Tlili et al., 2023). The integration is necessary worldwide, not only for the digitalization trend (Dergaa et al., 2023) but also as a support for students with the language barrier in academic writing and their efforts to learn as teachers in constructing teaching materials and assessment rubrics.

Artificial intelligence essay-writing tools help generate academic content and raise integrity issues, such as AI-assisted cheating (Bin-Nashwan et al., 2023). ChatGPT's potential to generate misinformation poses an academic threat (Tang et al., 2024), and its shortcomings—lack of emotional depth, fixed templates, and referencing issues—increase threats of plagiarism and academic dishonesty (Zou and Huang, 2023). Some educational institutions have prohibited ChatGPT, fearing its use will compromise the quality of students' writing, while others are allowing it, but with a caveat: learning how to use it effectively and responsibly. This two-pronged approach underscores the importance of balanced integration into academia (Yusuf et al., 2024).

Tools like ChatGPT can enhance potential people skills, such as grammar, vocabulary, and writing style, which can assist scientific writing (Dergaa et al., 2023). And, although tools like ChatGPT and Gemini accelerate processes such as checking for plagiarism and mining the literature, there's lingering skepticism about whether AI can draft full papers or deserve to be listed as an author. Even as these tools assist non-native writers, they threaten academic integrity, with risks including copyright violations, "artificial hallucinations" (false information), and biased outcomes. In addition, there are cases where AI-generated abstracts mislead human reviewers and raise concerns about whether AI can produce reliable scientific content (Tang et al., 2024).

Balancing the benefits of ChatGPT with ethical practices remains challenging for educators and policy-makers. Different countries emphasize different aspects of AI integration: the U.S. focuses on academic integrity and plagiarism (Kleebayoon and Wiwanitkit, 2023), the UK on equity in access (Roberts et al., 2023), and India on democratizing AI access despite security concerns (Taylor et al., 2023). Germany focuses on technical education (Von Garrel and Mayer, 2023), and Australia seeks to integrate ChatGPT alongside traditional teaching (Prem, 2019).

Theoretical foundation and research model

Technological, social, cultural, and psychological factors influence the introduction and adoption of

new technologies (Ivanov et al., 2024). The Technology Acceptance Model (TAM) (Davis, 1989), the Unified Theory of Acceptance and Use of Technology 2 (UTAUT2) (Venkatesh et al., 2012), and the Social Cognitive Theory (SCT) are key frameworks for analyzing technology acceptance, with TAM and UTAUT being particularly effective (Bouteraa et al., 2024). UTAUT, developed by Venkatesh et al. (2012), is known for its strong predictive power, making it suitable for technology diffusion (Strzelecki and ElArabawy, 2024). (Venkatesh et al., 2012) suggested that adding relevant factors could enhance UTAUT's effectiveness in specific contexts.

TAM is among the most utilized models in human-AI interaction studies, emphasizing new technology's psychological mechanisms (Acosta-Enriquez et al., 2024; Duong et al., 2023). However, TAM has been criticized for ignoring psychological states in immersive learning (Abdalla, 2024), where it may prove inadequate for the usage of AI (Zhang et al., 2023). SCT can promote a wider view considering personal, behavioral, and environmental predictors of technology acceptance, pointing out self-efficacy (Bandura, 1986) as one of the most important influences on motivation and behaviors. These theories together displayed a consolidated model for researcher usage of generative AI, by encompassing several determinants like perceived ease of use, perceived usefulness, facilitating conditions, social influence, risk, ethics, academic integrity, and self-efficacy (Figures 1 and 2).

Generative AI self-efficacy in academia

Self-efficacy, which is rooted in social cognitive theory (Bandura, 1986), is key to technology and information systems usage (Bin-Nashwan et al., 2023). It refers to an individual's confidence in their ability to successfully complete tasks utilizing a given technology. In the context of generative AI (GenAI), GenAI self-efficacy is described as individuals' confidence in their capacity to use GenAI tools, such as ChatGPT, for academic and research reasons (Wang et al., 2023). It has been studied in areas like the Internet (Torkzadeh and Van Dyke, 2001), ICT (Musharraf et al., 2018), healthcare robots (Turja et al., 2019), and AI (Shao et al., 2024; Wang and Chuang, 2024). Research suggests that AI self-efficacy differs from traditional technologies, especially in academia, where confidence in using tools like ChatGPT is vital. Although reactions to ChatGPT have been mixed, its potential in

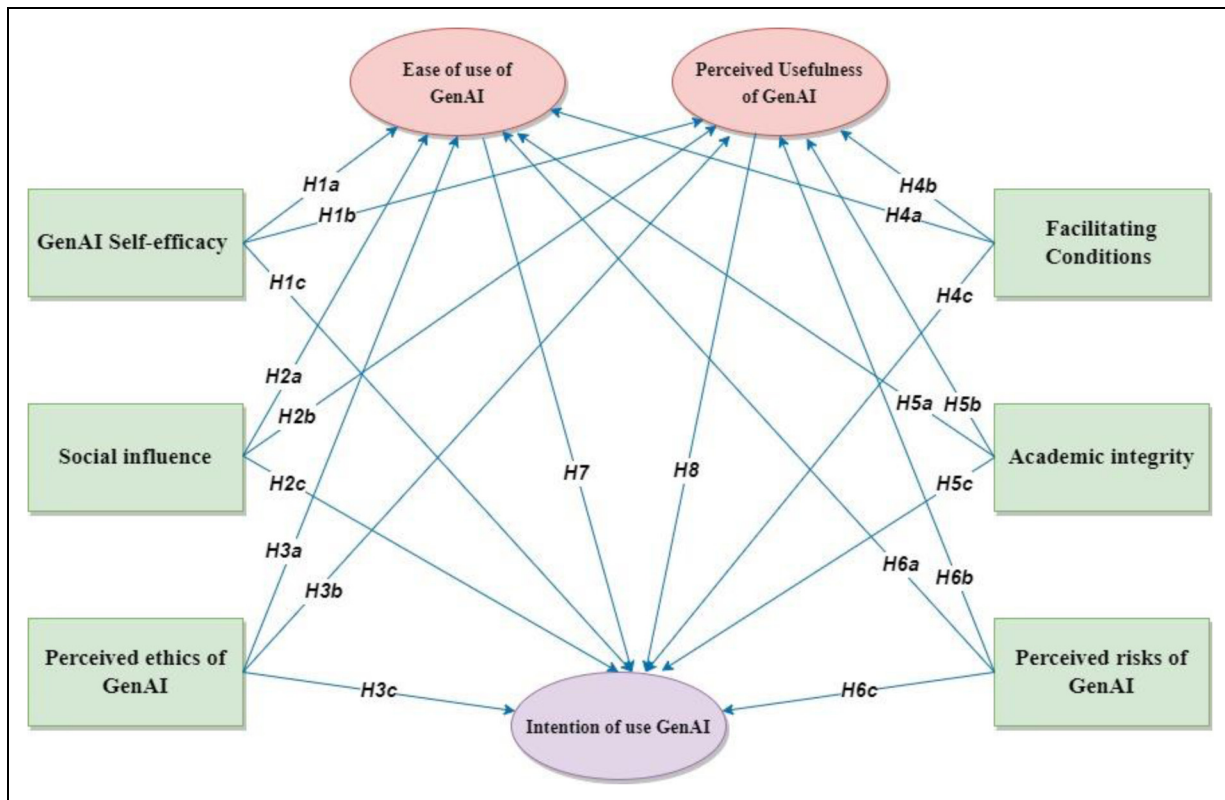


Figure 1. Conceptual model.

academia is widely recognized, aiding tasks like idea generation and article writing (Yusuf et al., 2024). High self-efficacy enhances perceived usefulness and ease of use (Hong, 2022). While robot self-efficacy affects AI's ease of use (AlGerafi et al., 2023), and academic self-efficacy positively influences AI adoption (Bin-Nashwan et al., 2023). Therefore, generative AI follows a similar acceptance pattern as previous technologies. Hypotheses have been proposed accordingly:

H1a. GenAI self-efficacy has a significant positive effect on ease of use,

H1b. GenAI self-efficacy has a significant positive effect on the perceived usefulness.

H1c. GenAI self-efficacy has a significant positive effect on the intention of use.

Social influence

Social influence from family, professors, and peers significantly impacts technology adoption (Venkatesh et al., 2003). Sohn and Kwon (2020) found that social

influence affects AI usage intentions in academia, with positive links to perceived usefulness, including for AI robots (AlGerafi et al., 2023). Studies by Al-Emran et al. (2023); Foroughi et al. (2024), and also support the positive correlation between social influence and chatbot or AI adoption, as seen among pre-service teachers (Zhang et al., 2023). However, the effect on perceived ease of use remains mixed, with some studies showing a positive effect (Abdalla, 2024), while others found no impact (Bouteraa et al., 2024). Positive social influence enhances generative AI's usefulness and ease of use, reducing risks and boosting confidence, while negative influence raises ethical concerns (Abdalla, 2024; Bouteraa et al., 2024; Menon and Shilpa, 2023; Strzelecki and ElArabawy, 2024). The conflicting results indicate that peer influence is context-dependent. Accordingly, the following hypotheses were derived from the aforementioned literature:

H2a. Social influence has a significant positive effect on ease of use.

H2b. Social influence has a significant positive effect on perceived usefulness.

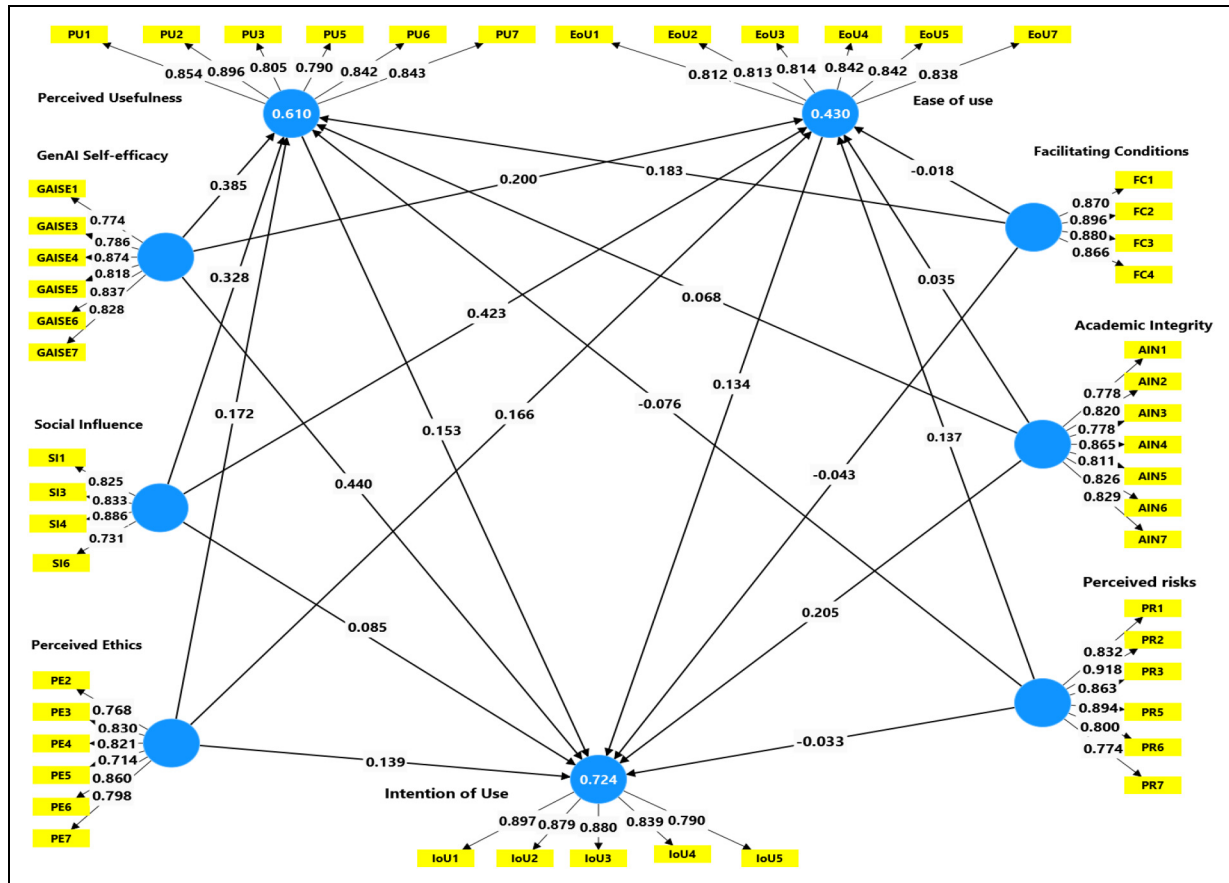


Figure 2. PLS Algorithm using SmartPLS.

H2c. Social influence has a significant positive effect on the intention of use.

Perceived ethics of generative AI

Perceived ethics in generative AI indicates perceptions the collective users may have of the ethical directions guiding these technologies' use: transparency, fairness, privacy, accountability, and equitability. It also reflects how far users believe a tool can progress in socially responsible applications without being biased and allowing respect for individual rights and societal norms (Shao et al., 2024). AI's adoption raises ethical concerns, but involving scholars and educators in AI-related decisions enhances autonomy and trust (Kieslich et al., 2024). Research on ChatGPT in academia shows positive and negative effects, with issues like academic dishonesty and technology overreliance on technology (Acosta-Enriquez et al., 2024). Ethical AI principles positively influence user acceptance and adoption intentions (Farhi et al., 2023). Jobin et al. (2019) highlighted 11 key AI

principles, including transparency, fairness, and privacy, with concerns about these principles impacting trust and adoption. Overreliance on AI may hinder creativity and critical thinking (Stahl and Eke, 2024), and ethical issues like plagiarism, bias, and misinformation have arisen in medical education (Sallam, 2023). Several studies suggest guidelines for AI's ethical use (Ivanov et al., 2024), noting that increased ethical concerns negatively impact adoption intentions (Etemad-Sajadi et al., 2022). However, alignment with ethical principles strengthens adoption intentions (Kieslich et al., 2024; Ko and Leem, 2021), enhancing perceived ease of use and usefulness (Farhi et al., 2023; Ko and Leem, 2021; Shao et al., 2024). As a result, based on the discussed literature, this study suggested the following hypothesis:

H3a. Perceived ethics have a significant positive effect on ease of use.

H3b. Perceived ethics have a significant positive effect on perceived usefulness.

H3c. Perceived ethics have a significant positive effect on the intention of use.

Facilitating conditions

Facilitating conditions in academia include access to instructions, technical support, and tool compatibility (Foroughi et al., 2024). Research on chatbots indicates that these factors significantly affect their adoption of them (Menon and Shilpa, 2023). The opposite is also true; they can drive the implementation of ChatGPT. Users who do not have access to the necessary technology or resources to use ChatGPT may be less inclined to adopt or continue using it (Strzelecki and ElArabawy, 2024). Conversely, resources and support foster adoption and integration into everyday tasks. Existing research has found that facilitating conditions, perceived usefulness, and ease of use contribute to the adoption of AI (Foroughi et al., 2024; Menon and Shilpa, 2023). Hence, the enabling factors are essential for understanding the overall perceived ease of use and use value of generative AI tools for research and intentionality. Based on this, the following hypotheses are proposed.

H4a. Facilitating conditions have a significant positive effect on ease of use.

H4b. Facilitating conditions have a significant positive effect on the perceived usefulness.

H4c. Facilitating conditions have a significant positive effect on the intention of use.

Academic integrity

Academic integrity—which entails honesty, respect, fairness, and responsibility in academic activities (Yusuf et al., 2024)—is essential to achieving academic excellence. The benefits of AI in education are that it increases engagement and promotes collaboration and accessibility; however, there are areas of concern, including integrity and plagiarism (Cotton et al., 2024). ChatGPT-type generative AI tools can improve self-confidence and efficiency or be used in academic misconduct (Eke, 2023; Gruenhagen et al., 2024). The fact that generative AI has sometimes proven unreliable produces inaccurate results, such as false references or conclusions (Lund et al., 2023; Sallam, 2023). This is critical when considering the ethical concerns and academic benefits AI brings. Within this context, the importance of exploring academic integrity and

its effects on generative AI use in academia applies to provide valuable information. While the direct impact of integrity on academics' behavior is well documented in literature (Bin-Nashwan et al., 2023; Cerda-Navarro et al., 2022; Tlili et al., 2023), few studies examine its role in shaping perceived ease of use and usefulness of generative AI in academia. Therefore, this study proposes hypotheses to explore these relationships:

H5a. Academic integrity has a significant positive effect on ease of use.

H5b. Academic integrity has a significant positive effect on the perceived usefulness.

H5c. Academic integrity has a significant positive effect on the intention of use.

Perceived risks

Perceived risk is considered one of the factors influencing decisions about adopting new technologies, including generative AI tools like ChatGPT (Sallam, 2023). Risks such as inaccurate content. Ethical concerns, such as bias, plagiarism, and copyright infringement; performance issues, privacy concerns; and threats to academic integrity can significantly impact user acceptance (Eke, 2023; Li, 2024). These risks are crucial in shaping user behavior and acceptance of AI tools in education. Studies using the TAM framework often treat perceived risks as a precursor to evaluating perceived usefulness and behavioral intentions (Mvondo et al., 2023; Wu et al., 2020). Research shows that perceived risks strongly influence the intention to adopt AI tools like ChatGPT (Abdaljaleel et al., 2024; Bhaskar et al., 2024; Li, 2024). Some studies report that these risks negatively affect perceived ease of use, usefulness, and adoption intention (Thakur and Srivastava, 2014). Recognizing the significance of these risks, this study aims to explore ethical concerns and risks associated with AI in academic settings. Based on this literature, the following hypotheses are proposed:

H6a. Perceived risks have a significant negative effect on the ease of use.

H6b. Perceived risks have a significant negative effect on the perceived usefulness.

H6c. Perceived risks have a significant negative effect on the intention of use.

Perceived usefulness and ease of use

The Technology Acceptance Model emphasizes perceived usefulness and ease of use as significant drivers of technology adoption. Perceived ease of use refers to how people perceive a system to perform and how much effort will be required to utilize it (Venkatesh et al., 2003). It plays a crucial role in adopting new technologies (Bhaskar et al., 2024). Perceived usefulness, which measures a technology's effectiveness in helping users achieve their goals, is also a strong predictor of usage intention. This is particularly evident with AI tools, where ease of use and usefulness strongly influence adoption (Wang et al., 2021). Generative AI tools like ChatGPT help users complete tasks efficiently, fostering positive perceptions of usefulness and increasing the intention to adopt these tools in research (Abdaljaleel et al., 2024; Abdalla, 2024; Acosta-Enriquez et al., 2024; Bhaskar et al., 2024; Shao et al., 2024). Based on this literature, the following hypotheses are proposed:

H7. Ease of use has a significant positive effect on the intention of use.

H8. Perceived usefulness of GenAI has a significant positive effect on the intention of use.

Research methodology

Research design and methodology

This study employed a quantitative research strategy, which is suited for examining models and causal relationships. The study employed a cross-sectional research approach to assess the GenAI tool usage model through a web-based survey distributed to Egypt's academic community. The online survey technology was primarily chosen for its efficient properties, such as responsiveness, cost-effectiveness, and availability to the target demographic (Fricker and Schonlau, 2002). Furthermore, the current research looks into technological challenges. This study's target sample consists of academic researchers from Egyptian universities.

Data collection

Using an anonymous online questionnaire, we used Google Forms to collect data between April and July 2024 (<https://forms.gle/oCVFo1sVcsFZN4fz5>). This period allowed for the minimum sample size to be secured. Non-probability sampling was used as the

total population size was unknown, given that the current study targeted faculty members from all Egyptian universities. The authors employed three sampling techniques to enhance the response rate and avoid non-response bias. First, convenience sampling was used to distribute the questionnaire link through emails. This method was selected due to its accessibility and practicality, allowing participants to be easily reached. However, additional methods were employed to address potential biases associated with this method, such as overrepresentation of specific demographics. Next, self-selection sampling was applied by posting the survey link on social media platforms, particularly within Egyptian research and academic groups with a significant proportion of university faculty members. This technique facilitated a broader reach, though it could introduce a bias towards individuals more interested in the study topic. To mitigate this, the researchers ensured the survey was widely shared across different groups. Finally, snowball sampling was applied by sharing the survey link with known individuals at various universities and requesting them to forward it to their networks. This helped extend the survey's reach to faculty members in various universities. The researchers ensured a diverse range of respondents to address potential biases in this method, such as overrepresenting certain networks. In total, 856 respondents completed the questionnaire, of which 97 were excluded for not using GenAI tools. Out of the remaining 759 responses, 17 were excluded for being incomplete. As a result, 742 respondents provided complete responses. The researchers examined these responses to guarantee the consistency of the questionnaire answers and validate their validity and usefulness for the study's future analysis.

Measurement tool

An online questionnaire was generated in March 2024. Before disseminating the questionnaire, we ran a pilot study to ensure its validity and reliability. A survey was conducted with 35 participants to assess the clarity of the data-gathering tool. The created scale accurately assesses faculty adoption of generative AI in higher education since all factors met validity and reliability criteria. The questionnaire was comprised of three sections. The initial part contained a filter question that segmented participants from those already using GenAI applications (e.g., Have you utilized a generative AI application in

your research/studies? Yes/No). Participants who had never used these tools could not complete the remaining sections. The second part collected demographic information from the study sample. The third part comprised 58 measurement items measuring nine concepts: generative AI self-efficacy, perceived ethics, academic integrity, social influence, facilitating conditions, perceived risks, ease of use, perceived usefulness, and intention to use GenAI applications in scientific research. Each measurement item was obtained from previously verified scales and modified to match the specific context of GenAI usage in academic contexts (The elements of each construct are available for review by clicking on the survey tool link in the appendix.). Participants' perceptions were assessed using a five-point Likert scale, ranging from one, indicating "strongly disagree," to five, indicating "strongly agree." Appendix 1 presents the statements and their sources.

Data analysis

SPSS 28 was used for statistical analysis of the socio-demographic data, and frequencies and percentages were presented. SmartPLS 4.1.0.7 is a strong statistical analysis software that has been used for over a decade to analyze data using partial least squares structural equation modeling (PLS-SEM) (Cheah et al., 2023). The ability of PLS-SEM, a variance-based modeling technique, to deal with small sample sizes, non-normal data distributions, and intricate

interactions between latent components has made it popular in the social sciences and management domains. PLS-SEM, as opposed to covariance-based SEM (CB-SEM), is especially well-suited for research projects that seek to forecast results rather than prove causality between constructs (Hair Jr et al., 2016b). It is widely applied in the higher education setting (Bin-Nashwan et al., 2023; Strzelecki and ElArabawy, 2024), making it an ideal tool to investigate the relationships hypothesized in this study.

Results

Sample profile

Table 1 illustrates the overall sample distribution, making the representative profiles of demographic, educational, and professional characteristics easy to grasp. The survey revealed a gender disparity (59.2% male, 40.8% female), with 52.2% aged 31–40; 40.8% were assistant professors, and 34.1% specialized in life sciences.

Common method bias (CMB)

Common method bias (CMB), a type of systematic error that results from the statistical association of the errors among variables measured with the same method, may inflate the strength of the relationship between variables. In accordance with the recommendations by Podsakoff et al. (2003) we carried out Harman's single-factor test which is based on the

Table 1. Demographics characteristics of participants (n = 742).

Demographics	Categories	N	%
Gender	Male	439	59.2
	Female	303	40.8
Age	20–30	131	17.7
	31–40	387	52.2
	41–50	178	23.9
	51+	46	6.2
	Demonstrator	51	6.9
Academic position	Lecturer/Instructor	158	21.3
	Postdoctoral Researcher/Fellow/Scholar	17	2.3
	Assistant Professor/Senior Lecturer	303	40.8
	Associate Professor	165	22.2
	Professor	48	6.5
Field of study/research	Social Sciences	114	15.4
	Life Sciences & Biomedicine	253	34.1
	Technology	177	23.8
	Physical Sciences	150	20.2
	Arts & Humanities	48	6.5

unrotated factor solution from an exploratory factor analysis conducted on all items of the study. The analysis results indicate that the variance explained by the first factor is 33.076%, which is lower than the 50% limit. To minimize CMB, we evaluated variance inflation factor (VIF) scores, with all values remaining below the acceptable threshold of 5 (Hair Jr et al., 2016b), suggesting no concern about CMB. These results further affirm the strength of our research design and findings.

Measurement model

This study evaluated the reflective model by assessing the variables' reliability and validity. Establishing requirements for internal consistency, discriminant validity, convergent validity, and dependability is critical. Based on Hair Jr et al. (2016b), a factor loading of more than 0.70 is regarded as appropriate for establishing reliability and validity. However, Hair Jr et al. (2016b), suggest that elements with loadings ranging from 0.40 to 0.70 be removed only if their absence improves composite dependability. In this study, eight items (GAISE2, SI2, SI5, PE1, PE8, PR4, EoU6, and PU4) must be eliminated from the original collection of 58 survey items. The average variance extracted (AVE) values exceeded the recommended threshold of 0.50, indicating convergent validity, as proposed by Hair. Table 2 shows that Cronbach's α values for all latent constructs above the required threshold of 0.70, demonstrating good internal consistency of the measures. While the composite reliability values exceeded the recommended threshold of 0.70, validating the internal consistency of the latent variables, further examination revealed room for improvement. An alternative reliability measure, ρ_A , developed by Dijkstra and Henseler (2015) accounts for differences in indicator correlations with more nuance. From complex sentences interspersed with shorter, simpler constructions emerges a more vivid narrative exploring the measures' reliability and validity. After all, precisely evaluating the instruments helps reduce errors and leads to more reliable conclusions, which remains the ultimate goal of any scholarly investigation.

The collinearity of the formative measurement models was assessed according to the VIF values shown in Table 2. All VIF scores of the indicators within the constructs have a value between 1.603 (for SI1) and 4.788 (for PU2). Since all the VIF values are far below the critical value of 5 (Hair Jr

et al., 2016a), multicollinearity is not a concern within any construct. This confirms that the indicators provide distinct and complementary contributions to their respective constructs, thereby supporting the validity of our measurement model.

Discriminant validity, a crucial component of the measurement model assessment, was assessed using the method put out by Fornell and Larcker (1981). The findings show that the latent constructs' square roots of average variance extracted (AVE) were higher than the correlation coefficients between them, as shown in Table 3. These findings demonstrate that each latent variable assesses a distinct construct that is not strongly associated with other factors, hence confirming discriminant validity. Furthermore, the discriminant validity was evaluated using the heterotrait-monotrait (HTMT) ratio, which measured the correlation between two items of a single construct. In keeping with the constructs' independence from one another, Table 4 demonstrates that all of the construct ratios were below the cautious cutoff of 0.85 (Kline, 2023).

Structural model

A structural model evaluation is performed to understand the effect of the exogenous constructs on the endogenous construct. The assessment consists of assessment collinearity between constructs, hypothesis testing, R^2 assessment, Q^2 assessment model predictive relevance, and effect size f^2 assessment. Bootstrapping was used to assess the significance of hypothesized relationships in 5000 draws. The analysis results based on the bootstrap method showed that all the path relationships of the model were significant ($t > \pm 1.65$, one-tailed test).

The analysis of bootstrapped path coefficients reveals that GenAI self-efficacy ($\beta = 0.200$, $p = 0.000$), social influence ($\beta = 0.423$, $p = 0.000$), and perceived ethics ($\beta = 0.166$, $p = 0.000$) positively impact the ease of use of GenAI in academia. Facilitating conditions and academic integrity have insignificant effects, while perceived risks show a significant positive relationship with ease of use ($\beta = 0.137$, $p = 0.001$). Hypotheses H1a, H2a, and H3a are supported, but H4a, H5a, and H6a are not.

Similarly, factors like GenAI self-efficacy ($\beta = 0.385$, $p = 0.000$), social influence ($\beta = 0.328$, $p = 0.000$), perceived ethics ($\beta = 0.172$, $p = 0.000$), facilitating conditions ($\beta = 0.183$, $p = 0.000$), and academic integrity ($\beta = 0.068$, $p = 0.011$) positively influence perceived usefulness. Perceived risks show a non-significant negative

Table 2. Descriptive statistics and measurement model results.

Construct / Items	Mean	St Dev.	Indicator loading	VIF	Cronbach's Alpha	Reliability coefficient	Composite reliability (CR)	Average variance extracted (AVE)
Generative AI Self-efficacy					0.902	0.905	0.925	0.673
GAISE1	3.73	0.897	0.774	2.321				
GAISE3	3.85	0.866	0.786	2.781				
GAISE4	3.98	0.760	0.874	3.449				
GAISE5	3.96	0.775	0.818	2.992				
GAISE6	4.10	0.817	0.837	2.983				
GAISE7	4.09	0.877	0.828	4.512				
Social Influence					0.842	0.880	0.892	0.674
SI1	3.84	1.092	0.825	1.603				
SI3	3.13	1.123	0.833	2.320				
SI4	3.44	0.931	0.886	2.847				
SI6	3.76	0.951	0.731	1.644				
Perceived Ethics					0.888	0.898	0.914	0.640
PE2	3.96	1.031	0.768	1.818				
PE3	4.34	0.859	0.830	3.426				
PE4	4.29	0.828	0.821	3.843				
PE5	4.31	0.876	0.714	1.858				
PE6	4.38	0.797	0.860	4.250				
PE7	4.34	0.764	0.798	3.299				
Facilitating Conditions					0.901	0.905	0.931	0.770
FC1	3.72	0.941	0.870	2.277				
FC2	3.57	0.942	0.896	3.231				
FC3	3.64	0.987	0.880	2.783				
FC4	3.65	0.930	0.866	2.426				
Academic Integrity					0.917	0.930	0.933	0.666
AIN1	3.54	1.020	0.778	2.234				
AIN2	3.65	0.943	0.820	2.281				
AIN3	3.79	0.995	0.778	3.811				
AIN4	3.86	0.851	0.865	3.276				
AIN5	3.84	1.009	0.811	3.963				
AIN6	3.76	1.009	0.826	3.099				
AIN7	3.84	0.976	0.829	2.799				
Perceived Risks					0.924	0.952	0.939	0.720
PR1	3.62	0.978	0.832	2.237				
PR2	3.61	0.983	0.918	4.337				
PR3	3.68	0.946	0.863	3.097				
PR5	3.65	0.973	0.894	3.418				
PR6	3.33	1.054	0.800	4.276				
PR7	3.30	1.036	0.774	3.750				
Ease of Use					0.908	0.911	0.928	0.684
EOU1	3.82	0.845	0.812	2.277				
EOU2	3.87	0.993	0.813	2.252				
EOU3	3.91	0.955	0.814	2.445				
EOU4	3.70	1.001	0.842	2.872				
EOU5	3.88	0.857	0.842	2.721				
EOU7	3.58	0.985	0.838	2.920				
Perceived Usefulness					0.916	0.917	0.934	0.704
PU1	4.01	0.883	0.854	3.651				
PU2	3.95	0.913	0.896	4.788				
PU3	3.87	0.887	0.805	2.431				
PU5	3.85	0.797	0.790	2.803				

(continued)

Table 2. (continued)

Construct / Items	Mean	St Dev.	Indicator loading	VIF	Cronbach's Alpha	Reliability coefficient	Composite reliability (CR)	Average variance extracted (AVE)
PU6	3.79	0.928	0.842	3.084	0.910	0.912	0.933	0.736
PU7	3.68	0.902	0.843	3.832				
Intention of Use								
IoU1	3.97	0.680	0.897	3.567				
IoU2	4.03	0.732	0.879	3.590				
IoU3	4.03	0.773	0.880	3.375				
IoU4	3.88	0.776	0.839	3.096	0.910	0.912	0.933	0.736
IoU5	4.40	0.688	0.790	2.253				

Table 3. Fornell-Larcker scale results.

Constructs	AIN	IoU	FC	GAISE	IoU	PE	PU	PR	SI
Academic Integrity	0.816								
Ease of use	0.262	0.827							
Facilitating Conditions	0.334	0.316	0.878						
GenAI Self-efficacy	0.341	0.500	0.346	0.820					
Intention of Use	0.487	0.569	0.313	0.773	0.858				
Perceived Ethics	0.379	0.388	0.284	0.585	0.599	0.800			
Perceived Usefulness	0.354	0.543	0.400	0.682	0.691	0.511	0.839		
Perceived risks	0.276	0.251	0.807	0.237	0.201	0.215	0.246	0.848	
Social Influence	0.153	0.540	0.227	0.390	0.455	0.161	0.551	0.084	0.821

Note: The values on the diagonal represent the square root of AVE.

Table 4. HTMT matrix.

Constructs	AIN	IoU	FC	GAISE	IoU	PE	PU	PR	SI
Academic Integrity									
Ease of use	0.275								
Facilitating Conditions	0.364	0.341							
GenAI Self-efficacy	0.362	0.536	0.381						
Intention of Use	0.513	0.609	0.338	0.850					
Perceived Ethics	0.401	0.408	0.325	0.631	0.642				
Perceived Usefulness	0.359	0.590	0.433	0.742	0.749	0.530			
Perceived risks	0.295	0.247	0.848	0.268	0.214	0.272	0.249		
Social Influence	0.183	0.588	0.250	0.415	0.483	0.163	0.599	0.112	

effect ($\beta = -0.076$, $p = 0.081$), supporting H1b, H2b, H3b, H4b, and H5b, but not H6b.

For the intention to use GenAI, self-efficacy ($\beta = 0.440$, $p = 0.000$), social influence ($\beta = 0.085$, $p = 0.000$), perceived ethics ($\beta = 0.139$, $p = 0.000$), and academic integrity ($\beta = 0.205$, $p = 0.000$) are positively significant, while facilitating conditions ($\beta = -0.043$, $p = 0.276$) and perceived risks ($\beta = -0.033$, $p = 0.402$) have insignificant negative effects. Ease of use ($\beta = 0.134$, $p = 0.000$) and perceived usefulness ($\beta = 0.153$, $p = 0.000$) also positively influence

intention, supporting H7 and H8. Hypotheses H1c, H2c, H3c, and H5c are supported, but H4c and H6c are not.

The model's explanatory power is demonstrated by the coefficient of determination (R^2) for the endogenous variables, which exceeds the recommended threshold of 0.02, indicating strong predictive accuracy. The model explains 43%, 61%, and 72.4% of the variance in ease of use, perceived usefulness, and intention to use GenAI in academia, respectively. Additionally, the PLSpredict procedure, used for out-

of-sample predictive power assessment, yielded Q^2 values greater than zero (0.420, 0.601, and 0.697), confirming the model's predictive relevance. The structural assessment results are summarized in Table 5, while the path coefficients in the structural model and the results of the significance tests is summarized in Table 6.

This study suggests that the construct dimensions (i.e., GenAI self-efficacy, perceived ethics, academic integrity, social influence, facilitating conditions, and perceived risks) affect the intentions of GenAI usage, mediated by ease of use and perceived usefulness. This mediating effect has not been extensively explored in the literature. Hence, we assessed the importance of these paths' indirect effects (Table 7).

Discussion and conclusions

Discussion

Generative AI offers benefits like content creation and collaboration in academia but raises concerns about academic integrity and plagiarism. This research integrates TAM, UTAUT2, and SCT with academic integrity and perceived ethics to create a theoretical framework, analyzing Egyptian academics' motivations for using GenAI.

The results of H1a, H1b, and H1c show that Generative AI (GenAI) self-efficacy significantly impacts individuals' responses in academic settings. High self-efficacy enhances perceived ease of use and usefulness, making GenAI seem less complex and more beneficial. This psychological factor encourages positive attitudes toward technology, reduces anxiety, and fosters exploration. As a result, GenAI self-efficacy increases perceived usefulness and ease of use, promoting the integration of these tools into academic routines for improved research quality and efficiency.

Furthermore, In line with prior literature (Hong, 2022), perceived usefulness and ease of use mediate the relationship between GenAI self-efficacy and intention to use. As users find the technology easy

and valuable, their likelihood of adoption increases. This study's findings align with research showing that higher technology self-efficacy leads to viewing GenAI as useful and easy to use (AlGerafi et al., 2023; Bin-Nashwan et al., 2023; Bouteraa et al., 2024; Hong, 2022; Shao et al., 2024). Institutions should enhance self-efficacy through training to integrate GenAI into academic research practices.

These results indicate that social influence affects Egyptian scholars' ease of use, perceived usefulness, and intention to use generative AI tools, thus supporting H2a, H2b, and H2c hypotheses. Finally, peer influence of both AI users (such as trainers and peers) and non-users (such as administrators) is important to the acceptance of AI, making the AI more perceived as helpful and easy to use and limiting perceived risks. Perceived usefulness and ease of use serve as an intermediate that relates social influence to intention to use. Past studies also provide similar trends with regard to social influence, ease of use, and perceived usefulness (Al-Emran et al., 2023; AlGerafi et al., 2023; Bouteraa et al., 2024; Menon and Shilpa, 2023; Zhang et al., 2023). However, negative social influence can raise concerns about AI's reliability and ethics, increasing resistance and worries about misuse, such as misinformation and biased content (Abdalla, 2024; Bin-Nashwan et al., 2023; Foroughi et al., 2024; Strzelecki and ElArabawy, 2024).

The current paper uncovered that researchers' perceptions regarding the ethics, usability, and applicability of generative AI instruments were strongly linked to their drive to embrace these resources, endorsing hypotheses H3a, H3b, and H3c. Ethical viewpoints are fundamental in adopting procedures and fostering trust in utilities like ChatGPT. Ease of use motivates acceptance by allowing scholars to smoothly navigate appliances, while perceived applicability inspires them by boosting productivity and efficiency. Usefulness and simplicity of use were found to facilitate the relationship involving perceived ethics and intention of use, demonstrating that ethics directly affect purpose and are reinforced through functional advantages. Thus, incorporating AI tools into scientific exploration necessitates attention to ethical elements, which enhance trust and mental comfort. Previous studies have established that ethical principles positively sway AI approval and adoption. This underscores the pivotal role of ethics in successfully integrating AI into investigational practices beyond only the technical facets (Acosta-Enriquez et al.,

Table 5. Explanatory power evaluation and predictive relevance of constructs.

Construct	R^2 value	R^2 adjusted	Q^2
Ease of use	0.430	0.425	0.420
Perceived Usefulness	0.610	0.606	0.601
Intention of Use	0.724	0.721	0.697

Table 6. Path coefficients in the structural model and the results of the significance tests.

Hypothesis	Path	Path coefficient	T-value	f2	P-value	Decision
H1a	GenAI self-efficacy → Ease of use	0.200	5.482	0.038	0.000	Supported
H1b	GenAI self-efficacy → Perceived usefulness	0.385	14.011	0.207	0.000	Supported
H1c	GenAI self-efficacy → Intention of use	0.440	14.354	0.311	0.000	Supported
H2a	Social influence → Ease of use	0.423	13.419	0.254	0.000	Supported
H2b	Social influence → Perceived usefulness	0.328	12.967	0.223	0.000	Supported
H2c	Social influence → Intention of use	0.085	3.910	0.015	0.000	Supported
H3a	Perceived ethics→ Ease of use	0.166	5.380	0.029	0.000	Supported
H3b	Perceived ethics→ Perceived usefulness	0.172	6.389	0.047	0.000	Supported
H3c	Perceived ethics→ Intention of use	0.139	5.829	0.040	0.000	Supported
H4a	Facilitating conditions → Ease of use	−0.018	0.415	0.000	0.678	Not Supported
H4b	Facilitating conditions → Perceived usefulness	0.183	4.270	0.026	0.000	Supported
H4c	Facilitating conditions → Intention of use	−0.043	1.089	0.002	0.276	Not Supported
H5a	Academic integrity → Ease of use	0.035	1.315	0.002	0.189	Not Supported
H5b	Academic integrity → Perceived usefulness	0.068	2.531	0.009	0.011	Supported
H5c	Academic integrity → Intention of use	0.205	8.060	0.119	0.000	Supported
H6a	Perceived risks→ Ease of use	0.137	3.278	0.011	0.001	Not Supported
H6b	Perceived risks→ Perceived usefulness	−0.076	1.744	0.005	0.081	Not Supported
H6c	Perceived risks→ Intention of use	−0.033	0.838	0.001	0.402	Not Supported
H7	Ease of use → Intention of use	0.134	4.537	0.037	0.000	Supported
H8	Perceived usefulness → Intention of use	0.153	5.742	0.033	0.000	Supported

Table 7. Specific indirect effects.

Path	Path coefficient	T-value	P-value
GenAI self-efficacy → Ease of use→ Intention of use	0.027	3.715	0.000
GenAI self-efficacy → Perceived usefulness→ Intention of use	0.059	5.547	0.000
Social influence → Ease of use→ Intention of use	0.057	4.794	0.000
Social influence → Perceived usefulness→ Intention of use	0.050	5.205	0.000
Perceived ethics→ Ease of use→ Intention of use	0.022	3.212	0.001
Perceived ethics→ Perceived usefulness→ Intention of use	0.026	3.857	0.000
Facilitating conditions → Ease of use→ Intention of use	−0.002	0.400	0.689
Facilitating conditions → Perceived usefulness→ Intention of use	0.028	3.230	0.001
Academic integrity → Ease of use→ Intention of use	0.005	1.279	0.201
Academic integrity → Perceived usefulness→ Intention of use	0.010	2.223	0.026
Perceived risks→ Ease of use→ Intention of use	0.018	2.910	0.004
Perceived risks→ Perceived usefulness→ Intention of use	−0.012	1.514	0.130

2024; Farhi et al., 2023; Ko and Leem, 2021; Shao et al., 2024).

Previous studies highlight those ethical concerns, including copyright issues, bias, plagiarism, inaccurate content, and misinformation, negatively affect users' intentions to adopt AI technologies (Etemad-Sajadi et al., 2022). To address this, academic institutions should establish policies on ethical AI usage, data security, and potential over-reliance (Lund et al., 2023; Mvondo et al., 2023). AI should be

viewed as a tool, not a replacement for human researchers, emphasizing the importance of human judgment in education. Future studies could explore whether educating users on AI ethics alleviates concerns and fosters a more positive adoption attitude.

Surprisingly, facilitating conditions had a negative impact on the ease of use and the intentions of Egyptian academics to adopt generative AI tools, leading to the rejection of hypotheses H4a and H4c. This negative effect may stem from technical

complexities or additional resource demands that exceed academics' expectations, lowering perceived ease of use. Furthermore, facilitating conditions (like infrastructure or training) might be inadequate or misaligned with academics' needs, reducing usage intentions as the tools may seem unready for practical implementation. Although improved system performance and efficiency positively influence perceived usefulness, as hypothesized in H4b, this effect diminishes once the technology problems blocking such improvements are resolved. It indicates that while generative AI tools can provide highly valued results, the absence of an easy-to-use experience may limit full adoption. Even with this Imperfect model, better and more user-friendly environments are required to gain the most of these tools across different locations. The supporting conditions are the most important for adopting chatbot programs in previous studies (Menon and Shilpa, 2023; Strzelecki and ElArabawy, 2024). Conversely, prior research on technology adoption has reported that facilitating conditions do not influence usage intention (Foroughi et al., 2024; Strzelecki and ElArabawy, 2024).

The results showed that academic integrity has no statistically significant effect on perceived ease of use ($\beta = 0.035$; $P > .05$), indicating no mediation. Nevertheless, the impact of academic integrity on perceived usefulness and Egyptian academics' intention to use generative AI supports Hypothesis H5b and Hypothesis H5c. One possible explanation for the lack of a significant effect of academic integrity on perceived ease of use lies in the conceptual distinction between ethical principles and the usability of technology. Academic integrity primarily revolves around adherence to ethical standards, such as honesty, fairness, and accountability in scholarly work. These principles, while crucial for responsible AI adoption, may not directly influence how users perceive the simplicity or complexity of interacting with generative AI tools. Ease of use is inherently tied to user-friendly interfaces, intuitive design, and technological accessibility, which are technical attributes independent of ethical considerations. Furthermore, for academics, the decision to adopt AI tools is likely shaped more by institutional policies and their relevance to academic tasks rather than individual assessments of technological usability through an ethical lens. This is consistent with what we know about the importance of ethical compliance to technology adoption within scholarly realms. Prior studies of academic integrity hold that proper student conduct is directly related to

instructors' actions (Cerdeña-Navarro et al., 2022; Cotton et al., 2024). However, some studies report a negative impact of academic integrity on AI adoption (Bin-Nashwan et al., 2023; Boutera et al., 2024; Dergaa et al., 2023; Eke, 2023; Yusuf et al., 2024). Other studies by Cotton et al. (2024) and Gruenhagen et al. (2024) revealed that academics require clear guidance on how to use tools such as GenAI in research and educational environments, highlighting the importance of establishing institutional policies and guidelines that ensure the responsible and ethical use of these tools in academic settings. In this context, the Supreme Council of Universities in Egypt has issued a guide titled "Regulations for the Use of Artificial Intelligence in Higher Education and Scientific Research."

Surprisingly, hypotheses H6a, H6b, and H6c are not supported, contrary to traditional technology adoption models like TAM, which emphasize the negative impact of perceived risk on perceived usefulness and intention to use (Bhaskar et al., 2024; Li, 2024; Sallam, 2023). This study shows no significant negative relationships. Interestingly, there is a positive association between perceived risk and ease of use ($\beta = 0.137$; $p = 0.001$), suggesting that as researchers become more aware of the risks associated with generative AI, they may adopt a more cautious but competent approach to its usage. The lack of a significant negative effect of perceived risk on perceived usefulness ($\beta = -0.076$; $P > .05$) challenges existing assumptions, possibly because researchers have already recognized AI's utility in academia, particularly in drafting, data analysis, and literature review. These concerns do not overshadow the perceived benefits, even with risks like plagiarism and bias. Similarly, the non-significant effect of perceived risk on intention to use ($\beta = -0.033$; $P > .05$) suggests that researchers' adoption decisions are driven more by pragmatic concerns—such as increasing productivity or staying competitive—than by perceived drawbacks. Egyptian researchers may view generative AI as essential to academia's evolution, making adoption nearly inevitable despite potential risks. These findings align with previous literature (Chen and Aklikokou, 2020; Ivanov et al., 2024).

Perceived usefulness and ease of use notably impact whether Egyptian scholars will embrace generative AI resources in their exploration. A user-friendly interface ($\beta = 0.134$, $p = 0.000$) proves pivotal for adoption since more straightforward tools demand less effort and cognitive function in academic

environments with time and brainpower constraints. Comparably, perceived usefulness ($\beta=0.153$, $p=0.000$) motivates scholars by highlighting perks like enhanced research efficiency. These findings align with the Technology Acceptance Model concepts, which stresses usability and simplicity as key determinants in scientists accepting novel methodologies. Furthermore, limited funding means researchers must stretch resources while maintaining productivity, incentivizing the selection of tools demanding less cost and work to implement. Studies by Abdalla (2024); AlGerafi et al. (2023); Bhaskar et al. (2024); Duong et al. (2023); Hong (2022); Ko and Leem (2021); Sohn and Kwon (2020); Zou and Huang (2023) have found comparable results, emphasizing that providing tangible benefits along with a user-friendly experience is essential in promoting the adoption of AI tools in academic research environments.

Theoretical implications

This study contributes to theoretical implications by enhancing insight regarding adopting AI tools such as ChatGPT in higher education institutions and bridging the gap in knowledge of factors mediating AI adoption by Egyptian researchers. It emphasizes constructs like self-efficacy, social impact, perceived morality, facilitating conditions, academic integrity, perceived risks, ease of use, and perceived usefulness. The study contributes to the literature by extending the existing models, such as the Technology Acceptance Model, to include some neglected elements, such as ethics and academic integrity. A particularly unusual and noteworthy finding is the role played by perceived risks, which, in fact, do not prevent the achieved adoption but can increase the perceived ease of use, suggesting that recognizing risk leads to users adopting risk-staving activities, adding complexity and insights into adoption models.

According to the study, the intention to use GenAI and its perceived usefulness are mainly influenced by ethics and integrity in the academic field. Ethical concerns such as excessive reliance on artificial intelligence and threats to the credibility of research emphasize the need for AI frameworks focused on ethics. It offers a validated model that combines social influences and facilitating conditions, providing an integrative framework for understanding how academia adopts synthetic intelligence. By examining the relationship between risk, ethics, and ease of use establishes a foundation for more comprehensive

theoretical frameworks that can address the ethical and practical implications of integrating artificial intelligence into academic research and suggests avenues for further investigation.

Practical implications

Due to the challenges of responsibly developing and deploying generative AI, the following recommendations could enhance the ability of institutions to implement principled but pragmatic policies. Key proposals include:

1. Universities should establish or propose clear regulations to govern the misuse of AI tools in research. These regulations are in line with international journal standards and specify penalties for violations.
2. Institutions must work closely with stakeholders, including journal editors and publishers, to devise effective methods to identify AI input in academic writing and adjust anti-spoofing policies accordingly.
3. Proposing clearly defined and acceptable mechanisms for the merit-based use of AI, such as providing technical assistance in academic writing, to avoid the overreliance on AI to generate substantial research content without human input review.
4. Universities should train their students on using AI responsibly and ethically—and warn them about the risks associated with plagiarized data—and incorporate AI ethics into existing research curricula to inform the next generation of researchers.
5. Specialized tools should be used to detect AI-generated texts, ensuring research integrity. Academic committees and theses should also be evaluated to prevent inappropriate AI involvement usage.
6. Universities should partner with scientific journals that reject AI-generated research and implement similar standards when assessing students' and researchers' work, while collaborating with global institutions to develop modern AI policies.
7. Mechanisms must be established to monitor research practices, ensuring researchers do not rely excessively on AI tools. Regular evaluations should verify adherence to technological and ethical standards advancements.

8. Specialized committees should be formed to review AI-related research. These committees must ensure compliance with ethical standards and retain independence to ensure impartiality in research evaluations.
9. AI developers, academic institutions, and publishers must work together to tackle unethical practices like plagiarism and fabricated citations, all while safeguarding researchers' rights rights.
10. As AI tools evolve, scientists and educators must actively engage in critical discussions and philosophical analyses of the implications, ethics, and drawbacks of integrating AI into academia.

Limitations and directions for future studies

Despite its contributions, this study has several restrictions, offering opportunities for future research. Relying solely on quantitative methods suggests integrating qualitative approaches for a more comprehensive understanding. The use of convenience sampling constrains the generalizability of results due to possible selection bias. One limitation of using an online survey is the potential for response bias, as it may overrepresent certain demographics and individuals more interested in the topic. Including students would strengthen the robustness of future research. Exploring the use of generative AI across countries could reveal distinct cultural, social, and economic patterns. Moreover, future research should explore the technical complexities and resource demands causing the negative impact of facilitating conditions on generative AI adoption. Future studies could enrich existing theoretical models like TAM, UTAUT2, and SCT by merging theories such as Diffusion of Innovations and incorporating variables like stress, trust, past AI experience, social presence, time savings, and electronic word-of-mouth. This study further neglected moderating impacts of participant traits, for instance, gender, age, academic discipline, and degree, on technology adoption. Ultimately, potential research should expand ethical considerations by assessing factors like privacy, fairness, transparency, accountability, and bias, which are subjective and require more detailed investigation (Ko and Leem, 2021).

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Appendix I. Measurement scale. (<https://is.gd/QnpIFC>)